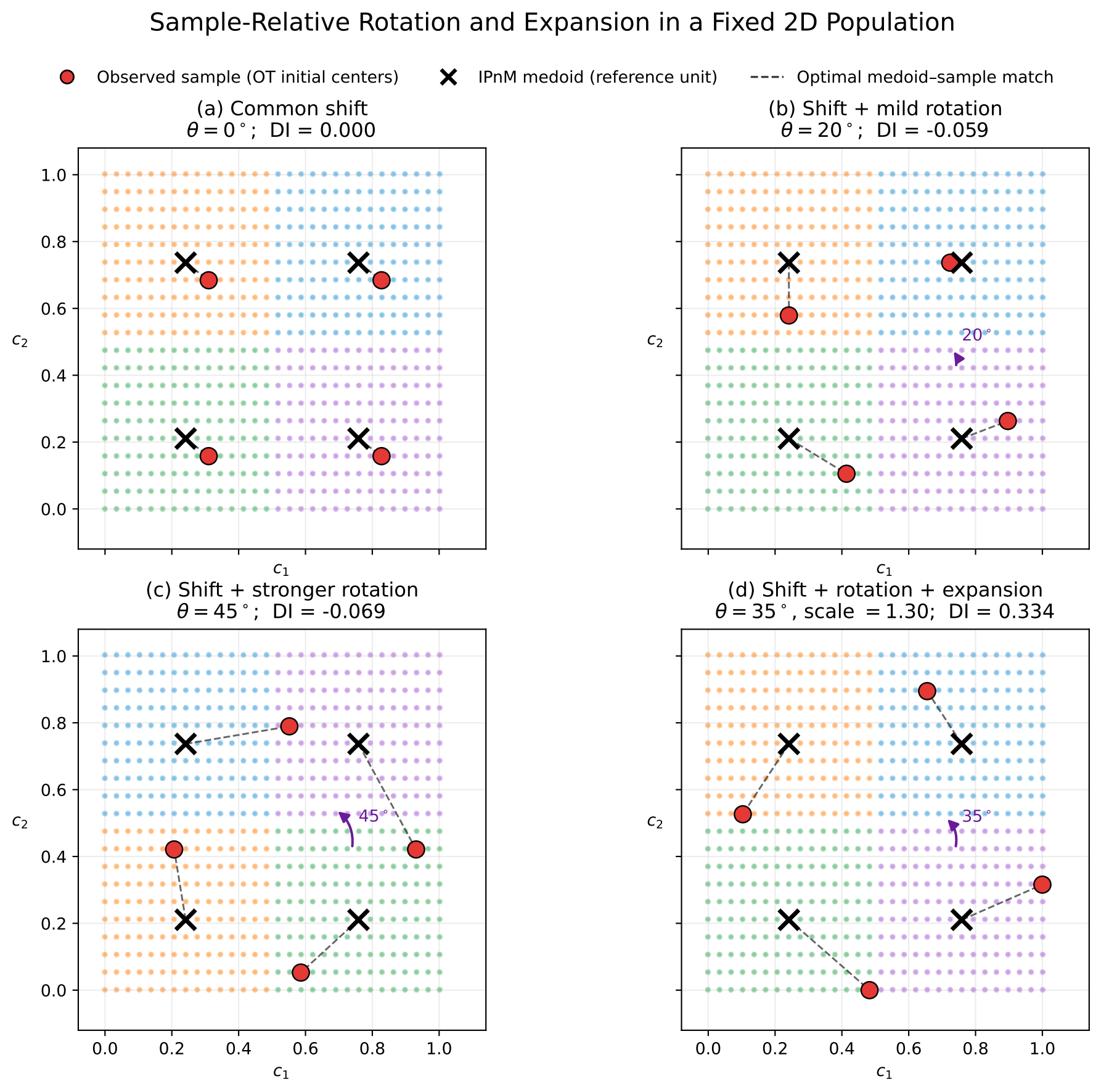

Sample-Relative Rotation and Expansion in a Fixed 2D Population
Observed sample (OT initial centers)
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Common shift
θ = 0°; DI = 0.000
(b) Shift + mild rotation
θ = 20°; DI = -0.059
20°
(c) Shift + stronger rotation
θ = 45°; DI = -0.069
45°
(d) Shift + rotation + expansion
θ = 35°, scale = 1.30; DI = 0.334
35°
c1
c2
0.0
0.2
0.4
0.6
0.8
1.0